UNIT-1

1. **Array implementation of LIST ADT (16)**
2. **Linked list implementation of LIST ADT (16)**
3. **Circular Linked list (16)**
4. **Doubly Linked list (16)**
5. **Application of List- Polynomial ADT (16)**

UNIT-2

1. **Stack ADT Array implementation (16)**
2. **Stack ADT Linked list implementation (16)**
3. **Queue ADT Array implementation (16)**
4. **Queue ADT Linked list implementation (16)**
5. **Circular queue (16)**
6. Application of Stack (Balancing symbols, **Infix to postfix conversion, Post fix expression Evaluation**, Function calls) Any problem can be asked for 8 marks

UNIT-3

1. **Tree Traversals (8)**
2. Binary Tree ADT (8)
3. Expression trees (8)
4. **Binary Search Tree ADT (16)**
5. **AVL Trees (16)**
6. **Priority Queue (Heaps) & Binary Heap.(16)**

UNIT-4

1. **B-Tree (8)**
2. B+ Tree (8)
3. **Breadth-first traversal (8)**
4. **Depth-first traversal (8)**
5. **Euler circuits (16)**
6. Topological Sort (16)
7. **Dijkstra's algorithm (16)**

8. Minimum Spanning Tree – Prim's algorithm – Kruskal's algorithm

(16)UNIT-5

1. Searching – Linear Search & Binary Search. (16) 2. Bubble sort (8)

3. Selection sort (8)

4. Insertion sort (8)

5. Shell sort (8)

6. Merge Sort (8)

7. **Hashing –Separate Chaining, Open Addressing**, Rehashing, Extendible Hashing. (16)